

## AI ETHICS LEARNING TOOLKIT

# Is Ai Biased?

## Exploring Bias and Monoculture

**“Algorithmic oppression is not just a glitch in the system but, rather, is fundamental to the operating system of the web.”**

**– Dr. Safiya Noble, Information Sciences Scholar**

The concept of “garbage in, garbage out” illustrates a core aspect of AI’s limitations: biased training data produces biased outputs. The exact training datasets used by models like GPT-4 are kept secret, but we know they rely on massive collections of human-generated text scraped from the open internet, including sites like Reddit, Wikipedia, and countless websites and forums. In a famous critique on the ethics of Large Language Models, researchers note that the internet is overrun with hegemonic or monocultural viewpoints which leads to an overrepresentation of white supremacist, misogynistic, and ageist perspectives in the training data. As AI usage grows, these biased worldviews are encoded in its data and potentially amplified into the culture. One good example of this bias is

linguistic diversity. There are over 7,100 languages spoken in the world, yet very few of those languages are documented on the Internet, and even fewer are supported by Generative AI models. Given the risks of AI bias, students should reflect on the many points at which bias can enter the AI pipeline—from what data is included, to how it's labeled or filtered, to what questions the users ask. Understanding these potential areas of bias helps students critically evaluate the information AI provides—and consider who gets represented, and who gets left out.

## Learning Activities

### **Conversation Starters** A Few Questions to Get the Discussion Going

- What types of bias have you noticed on social media, or the internet more broadly? Did these shape your opinion or behavior?
- Have you ever noticed any biases, or inaccuracies, in AI-generated responses? What made you question it?
- What strategies do you use to spot bias in news, social media, or AI-generated content?
- How might generative AI unintentionally reinforce harmful stereotypes or deepen social inequalities? Can you think of examples?
- Whose perspectives do you think are most represented in online content and AI outputs? Whose voices are left out – and why might that be?



## Active Learning with AI

### Fun Ways to Explore AI's Strengths and Limitations

- Prompt an AI image generator for an image of a lawyer, engineer, doctor, etc, and compare the gender, ethnicity, age of the person. Were there stereotypes represented in the outputs?
- Prompt AI for a brief history of an historical event or social phenomenon that the class is studying. Discuss the perspectives/biases present and absent in the AI's output. Examine what sources it used (if available).
- **No AI Alternative:** Don't use an image AI generator, but check out this article on [AI stereotypes](#) in which authors explored images and stereotypes with AI. Discuss your reactions in small groups.



## Disciplinary Extensions

### Ideas for Exploring AI's Impact in Specific Fields

- **Linguistics course:** Students can explore how AI handles sociolinguistics and code-switching. For example, students could prompt AI with different dialects, or registers, and see how it understands these and responds. Does it try to "correct"/normalize the prompting language?
- **Social Science courses (ex. GSF, AMES, AAAS):** Students can use AI to explore how it frames certain topics or identities. Connect this activity to theories and readings discussed in class about stereotypes and cultural biases/discrimination.  **TRIGGER WARNING:** Instructors should take care to frame an AI activity thoughtfully, as the AI outputs may be unexpected and/or triggering.
- **Foreign Language course:** Students could prompt AI for a story (in foreign language) that includes potential for biased outputs. Example: Help me write a story about a doctor and a nurse

working in a small, rural village. Does it gender the doctor/nurse?  
Does it include rural stereotypes?

## Resources

- Perrigo, B. (2022, January 18). [Why Timnit Gebru Isn't Waiting for Big Tech to Fix AI's Problems](#). *TIME*. [Magazine article]  
- Busiek, J. (2024, March 21). [Three fixes for AI's bias problem](#). *University of California*. [Blog post] 
- Nicoletti, L., & Equality, D. B. T. +. (2025, January 17). [Humans are biased. Generative AI is even worse](#). *Bloomberg*. [News article] 
- Turk, V. (2023, October 10). [How AI reduces the world to stereotypes](#). *Rest of World*. [Article] 
- Jeffries, A. (2017, April 27). [Machine learning is racist because the internet is racist](#). *The Outline*. [Article] 
- Lizarraga, L. (2023, November 8). [How does a computer discriminate? Interview with Safiya Noble](#). Code Switch Podcast. *NPR*. [Podcast] 

## Scholarly

- Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). [On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?](#) . *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. [Article] 
- Gebru, T., & Torres, É. P. (2024). [The TESCREAL bundle: Eugenics and the promise of utopia through artificial general intelligence](#). *First Monday*. [Journal article] 
- Noble, S. U. (2018). [Algorithms of Oppression: How Search Engines Reinforce Racism](#). NYU Press. [Book – available in Duke Libraries]  
- Sourati, Z., Karimi-Malekabadi, F., Ozcan, M., McDaniel, C., Ziabari, A., Trager, J., Tak, A., Chen, M., Morstatter, F., & Dehghani, M. (2025). [The Shrinking Landscape of Linguistic Diversity in the Age of Large Language Models](#). *arXiv*. [Preprint] 
- Hofmann, V., Kalluri, P.R., Jurafsky, D. et al. [AI generates covertly racist decisions about people based on their dialect](#). *Nature* 633, 147–154 (2024). [Journal article]  

## Recommendations

- Related topics → Does AI spread mis/disinfo? Does AI harm critical thinking?
  - [AI Pedagogy Project \(Harvard\) Assignments](#) → Filter by theme (e.g. bias) and/or subject (e.g. ethics & philosophy)
  - [Bias-related Articles](#) from the AI Ethics & Policy News Aggregator sourced by [Casey Fiesler](#). Note: This would be an excellent place to identify recent news stories you could share with students, or incorporate into a case study.
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1. Noble, S. U. (2018). *Algorithms of Oppression: How Search Engines Reinforce Racism*. NYU Press.
2. Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the Dangers of Stochastic Parrots: Can Language Models Be Too Big? *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, 610–623. <https://doi.org/10.1145/3442188.3445922>

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